The Gasoline System

The Carburetor—how does it work?

Answer No. 42

The carburetor is of the automatic float feed type, having but one adjustment—the gasoline needle valve. The cross-section diagram of carburetor on page 20 shows how the gasoline enters the carburetor, is vaporized by a current of air and passes through the inlet pipe to the engine in the form of an explosive mixture. The gasoline, entering the bowl of the carburetor, gradually raises the float to a point where the inlet needle is forced upwards into its seat, thus cutting off the flow of gasoline. As the gasoline in the bowl recedes, the float lowers, allowing the needle to drop from its seat and the flow of gasoline is resumed. It is plain to see that a constant level of gasoline is maintained in the carburetor by the automatic action of float and needle. The quantity of gasoline entering into the mixture is governed by the needle valve (see Answer No. 45). The volume of gas mixture entering the intake pipe is controlled by opening and closing the throttle, according to the speed desired by the driver.

Why is Carburetor adjustment placed on dash?

Answer No. 43

For the convenience of the driver in adjusting the carburetor. After the new car has become thoroughly worked in, the driver should observe the angle of the carburetor adjusting rod at which the engine runs most satisfactorily. In cold weather it will probably be found necessary to turn the dash adjustment one-quarter turn to the left, particularly in starting a cold engine. As gasoline vaporizes readily in warm weather, the driver will find it economical to reduce the quantity of gasoline in the mixture by turning the carburetor adjustment to the right as far as possible without reducing speed. This is particularly true when taking long drives where conditions permit a fair rate of speed being maintained, and accounts for the excellent gasoline mileage obtained by good drivers.

What is meant by a "lean" and a "rich" Mixture?

Answer No. 44

A lean mixture has too much air and not enough gasoline. A rich mixture has too much gasoline and not enough air. A rich mixture will not only quickly cover the cylinders, pistons and valves with carbon, but will tend to overheat the cylinders, and is likewise wasteful of the fuel. It will often choke the engine and cause misfiring at slow speeds, although at high speeds the machine will run perfectly. The mixture should be kept as lean as possible without the sacrifice of any of the power of the motor. A lean mixture will often result in backfiring through the carburetor, for the reason that the gas burns slowly in the cylinder, and is still burning when the inlet valve opens again, which causes the gas in the intake to ignite. A rich mixture is shown by heavy, black exhaust smoke with a disagreeable smell. Proper mixture will cause very little smoke or odor.

How is the Carburetor adjusted?

Answer No. 45

The usual method of regulating the carburetor is to start the motor, advancing the throttle lever to about the sixth notch, with the spark retarded to about the fourth notch. The flow of gasoline should now be cut off by screwing the needle valve down to the right until the engine begins to misfire, then gradually increase the gasoline feed by opening the needle valve until the motor picks up and reaches its highest speed and no trace of black smoke comes from the exhaust. Whenever it is necessary to turn the adjusting needle down more than a quarter turn below

its normal position, the lock nut on top of the carburetor at the point through which needle passes should first be loosened, as otherwise it is impossible to tell when the needle is turned down in its seat too far. Turning the needle down too tightly will result in its becoming grooved and the seat enlarged. When these parts are damaged it is difficult to maintain proper adjustment of the carburetor. Having determined the point where the motor runs at its maximum speed, the needle valve lock nut should be tightened to prevent the adjustment being disturbed. For average running, a lean mixture will give better results than a rich one.

Why does water clog the Carburetor? **Answer No. 46**

The presence of water in the carburetor or gasoline tank, even in small amounts, will prevent easy starting and the motor will misfire and stop. As water is heavier than gasoline it settles to the bottom of the tank and into the sediment bulb along with other foreign matter. As it is difficult nowadays to get gasoline absolutely free from impurities, especially water, it is advisable to frequently drain the sediment bulb under the gasoline tank. During cold weather the water which accumulates in the sediment bulb is likely to freeze and prevent the flow of gasoline through the pipe leading to the carburetor. Should anything of this kind happen it is possible to open the gasoline line by wrapping a cloth around the sediment bulb and keeping it saturated with hot water for a short time. Then the water should be drained off. In event the water gets down into the carburetor and freezes, the same treatment may be applied.

What makes the Carburetor leak? **Answer No. 47**

The flow of gasoline entering the carburetor through the feed pipe is automatically regulated by the float needle raising and lowering in its seat. Should any particle of dirt become lodged in the seat, which prevents the needle from closing, the gasoline will overflow in the bowl of the carburetor and leak out upon the ground.

When there is dirt in the Carburetor—what? **Answer No. 48**

The spraying nozzle of the carburetor having a very small opening, a minute particle of grit or other foreign matter will clog up the orifice, and result: motor will begin to misfire and slow down as soon as it has attained any considerable speed. This is accounted for by the fact that at high speeds the increased suction will draw the particles of dust, etc., into the nozzle. By opening the valve needle half a turn and giving the throttle lever two or three quick pulls the dirt or sediment will often be drawn through, when the needle may be turned back to its original place. If this does not accomplish the purpose, the carburetor should be drained.

If Engine runs too fast or chokes with throttle retarded—what? **Answer No. 49**

If the engine runs too fast with throttle fully retarded unscrew the carburetor throttle lever adjusting screw until the engine idles at suitable speed. If the motor chokes and stops when throttle is fully retarded the adjusting screw should be screwed in until it strikes the boss, preventing the throttle from closing too far. When proper adjustment has been made, tighten lock-screw so that adjustment will not be disturbed.

What is the purpose of the Hot Air Pipe? **Answer No. 50**

It takes the hot air from around the exhaust pipe and conducts it to the carburetor—where the heat facilitates the vaporising of the gasoline. It is usually advis-

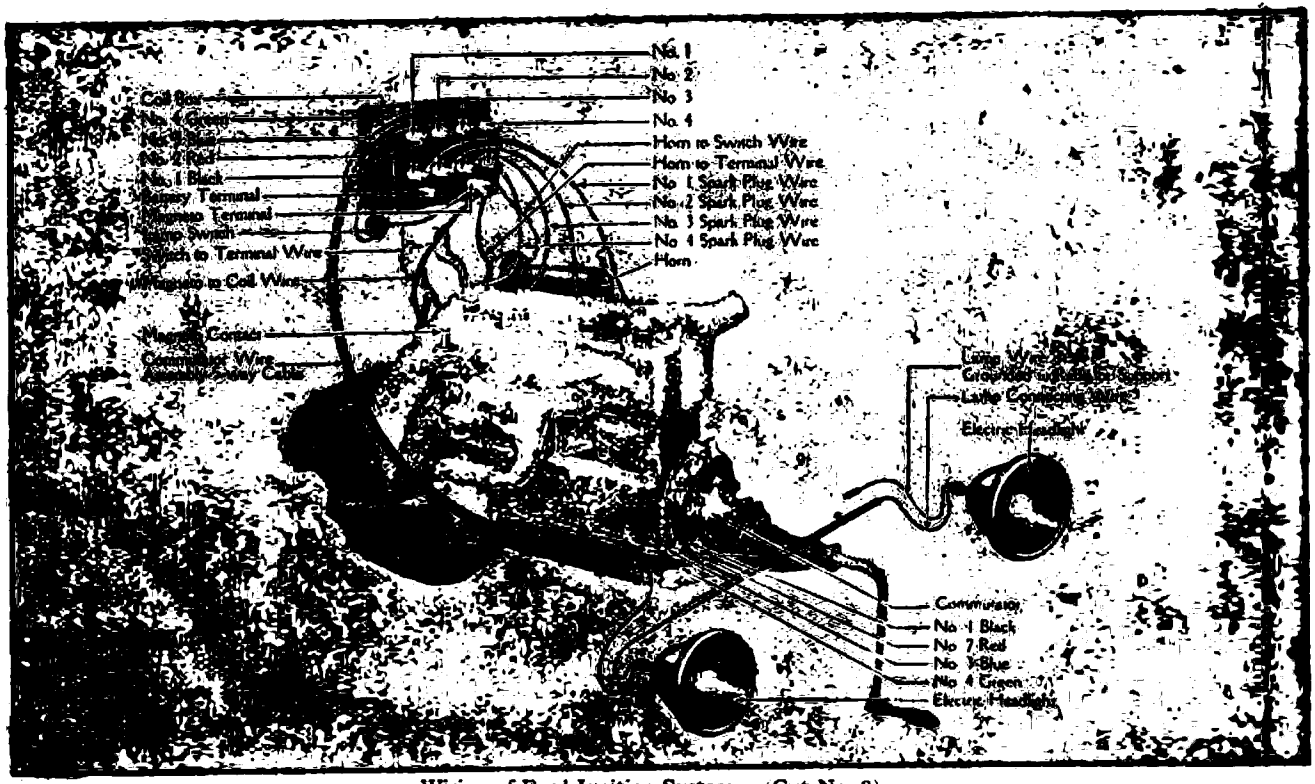
able to remove this pipe in the hot season—but it is an absolutely necessary feature during cold weather.

What is the purpose of the Cork Float? **Answer No. 51**

It automatically controls the flow of the gasoline into the carburetor. If it floats too low, starting will be difficult; if too high, the carburetor will flood and leak. A cork float which has become fuel soaked should be removed and replaced by a new one or thoroughly dried and then given a couple of coats of liquid shellac to make it waterproof.

Should Priming Rod be used in starting when Motor is warm? **Answer No. 52**

No. The carburetor does not ordinarily require priming when the motor is warm, and starting with the rod pulled out is apt to "flood" the engine with an over-rich mixture of gas, which does not readily explode. This naturally causes difficulty in starting. If you should accidentally flood the engine, turn the carburetor adjusting needle down (to the right) until it seats; then turn the engine over a few times in order to exhaust the rich gas. As soon as the motor starts, turn back the needle (to the left) and readjust the carburetor.



Wiring of Ford Ignition System. (Cut No. 8)

The Ford Ignition System

What is the purpose of the Ignition System? **Answer No. 53**

It furnishes the electric spark which explodes the charge in the combustion chamber, thus producing the power which runs the engine. It is important that the charge be correctly ignited at the proper time, in order to obtain satisfactory results in running the car. In the Ford car the ignition system is as simple as it is possible for human invention to make it.

How does the Magneto generate the current? **Answer No. 54**

In revolving at the same rate of speed as the motor, the magnets on the flywheel passing the stationary coil spools create an alternating low tension electric current in coils of wire which are wound around spools fastened to the stationary part of the magneto, and is carried from these coils to the magneto connection (wire) leading to the coil box on the dash.

Should the Coil Vibrator Adjustment be disturbed? **Answer No. 55**

The present style of coil unit is properly adjusted when it leaves the factory and this adjustment should not be disturbed unless to install new points or to reduce the gap between the points which may have increased from wear. When adjustments are necessary they should, whenever possible, be made by one of our service stations who have special equipment for testing and adjusting units and will gladly furnish expert service. If the points are pitted they should be filed flat with a fine double-faced file and the adjusting thumb nut turned down so that with the spring held down the gap between the points will be a trifle less than $\frac{1}{32}$" of an inch. Then set the lock nut so that the adjustment can not be disturbed. Do not bend or hammer on the vibrators, as this would affect the operation of the cushion spring of the vibrator bridge and reduce the efficiency of the unit.

How is a Weak Unit detected? **Answer No. 56**

With the vibrators properly adjusted, if any particular cylinder fails or seems to develop only a weak action, change the position of the unit to determine if the fault is actually in the unit. The first symptom of a defective unit is the buzzing of the vibrator with no spark at the plug. Remember that a loose wire connection, faulty spark plug, or worn commutator may cause irregularity in the running of the motor. These are points that should be considered before laying the blame on the coil.

How may short circuit in Commutator Wiring be detected? **Answer No. 57**

Should the insulation of the primary wires (running from coil to commutator) become worn to such an extent that the copper wire is exposed—the current will leak out (i. e. short circuit) whenever contact with the engine pan or other metal parts is made. A steady buzzing of one of the coil units will indicate a "short" in the wiring. When driving the car the engine will suddenly lag and pound on account of the premature explosion. Be careful not to crank the engine downward against compression when the car is in this condition, as the "short" is apt to cause a vigorous kick back.

Does Coil Adjustment affect starting? *Answer No. 58*

Yes. When the vibrators are not properly adjusted more current is required to make and break the contact between the points, and, as a result, at cranking speeds you would not get a spark between the spark plug points. Do not allow the contact points to become "ragged," otherwise they are apt to stick and cause unnecessary difficulty in starting, and when running they are apt to produce an occasional "miss" in the engine.

What is the purpose of the Commutator? *Answer No. 59*

The commutator (or timer) determines the instant at which the spark plugs must fire. It effects the "make and break" in the primary circuit. The grounded wire in the magneto allows the current to flow through the metal parts to the metal roller in the commutator. Therefore, when the commutator roller in revolving touches the four commutator contact points, to each of which is attached a wire

The Ford Commutator. (Cut No. 9)

connected with a coil unit, an electrical circuit is passed through the entire system of primary wires. This circuit is only momentary, however, as the roller passes over the contact point very rapidly and sets up the circuit in each unit as the roller touches the contact point connected with that unit. The commutator should be kept clean and well oiled at all times.

What about the Spark Plugs? *Answer No. 60*

One is located at the top of each cylinder and can easily be taken out with the spark plug wrench included with every car, after the wire connection is removed. The high voltage current flows out of the secondary coils in the coil box and on reaching the contact points in each spark plug it is forced to jump a $\frac{1}{32}''$ gap, thereby forming a spark which ignites the gasoline charge in the cylinders.

The spark plugs should be kept clean (i. e., free from carbon) and should be replaced if they persist in not working properly. There is nothing to be gained by experimenting with different makes of plugs. The make of plugs with which Ford engines are equipped when they leave the factory are best adapted to the requirements of our motor, notwithstanding the opinion of various garage men to the

contrary. All wire connections to spark plugs, coil box and commutator should, of course, at all times be kept in perfect contact.

What are the indications of Ignition trouble? **Answer No. 61**

The uneven sputter and bang of the exhaust means that one or more cylinders are exploding irregularly or not at all, and that the trouble should be promptly located and overcome. Misfiring, if allowed to continue, will in time injure the engine and the entire mechanism. If you would be known as a good driver you will be satisfied only with a soft, steady purr from the exhaust. If anything goes wrong, stop and fix it if possible—don't wait until you get home.

How can one tell which Cylinder is missing? **Answer No. 62**

This is done by manipulating the vibrators on the spark coils. Open the throttle until the engine is running at a good speed and then hold down the two outside vibrators, No. 1 and No. 4, with the fingers, so they cannot buzz. This cuts out the two corresponding cylinders, No. 1 and No. 4, leaving only No. 2 and No. 3 running. If they explode regularly it is obvious the trouble is in either No. 1 or No. 4. Relieve No. 4 and hold down No. 2 and No. 3 and also No. 1; if No. 4 cylinder explodes evenly it is evident the misfiring is in No. 1. In this manner all of the cylinders in turn can be tested until the trouble is located. Examine both the spark plug and the vibrator of the missing cylinder.

If the Coil and Plug are right—what? **Answer No. 63**

The trouble is probably due to an improperly seated valve, worn commutator, or short circuit in the commutator wiring. Weakness in the valves may be easily determined by lifting the starting crank slowly the length of the stroke of each cylinder in turn, a strong or weak compression in any particular valve being easily detected. It sometimes happens that the cylinder head gasket (packing) becomes leaky—permitting the gas under compression to escape, a condition that can be detected by running a little lubricating oil around the edge of the gasket and noticing whether bubbles appear or not.

Does a worn Commutator ever cause misfiring? **Answer No. 64**

Yes. If misfiring occurs when running at high speed, inspect the commutator. The surface of the circle around which roller (see Cut No. 9) travels should be clean and smooth, so that the roller makes a perfect contact at all points. If the roller fails to make a good contact on any one of the four contact points, its corresponding cylinder will not fire. Clean these surfaces, if dirty. In case the fibre, contact points and roller of the commutator are badly worn the most satisfactory remedy is to replace them with new parts. The spring should be strong enough to make a firm contact between the roller points if they are worn or dirty.

Misfiring may also be caused by short circuited commutator wires.

How is the Commutator removed? **Answer No. 65**

Remove cotter pin from spark rod and detach latter from commutator. Loosen the cap screw which goes through breather pipe on top of time gear cover. This will release the spring which holds the commutator case in place and this part can be readily removed. Unscrew lock nut; withdraw steel brush cap and drive out the retaining pin. The brush can then be removed from the cam shaft.

In replacing the brush, care must be exercised to see that it is replaced so that the exhaust valve on the first cylinder is closed when the brush points upward. This may be ascertained by removing the valve door and observing the operation of No. 1 valve.

Does cold weather affect the Commutator?

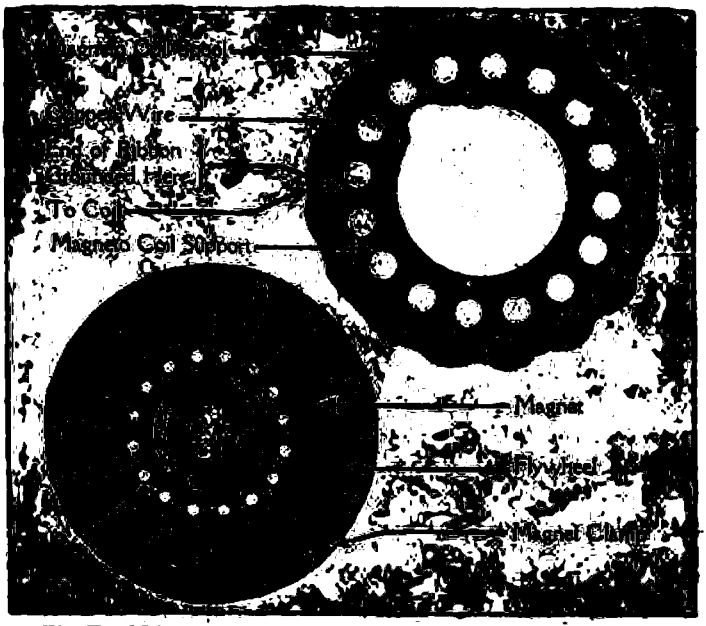
Answer No. 66

It is a well-known fact that in cold weather even the best grades of lubricating oil are apt to congeal to some extent. If this occurs in the commutator it is very apt to prevent the roller from making perfect contact with the contact points imbedded in the fibre. This, of course, makes difficult starting as the roller arm spring is not stiff enough to brush away the film of oil which naturally forms over the contact points. To overcome this, as well as any liability of the contact points to rust, we recommend a mixture of 25% kerosene with commutator lubricating oil, which will thin it sufficiently to prevent congealing, or freezing, as it is commonly called. You have probably noticed in starting your car in cold weather that perhaps only one or two cylinders will fire for the first minute or so, which indicates that the timer is in the condition described above and as a consequence a perfect contact is not being made on each of the four terminals.

How is the Magneto removed?

Answer No. 67

It is necessary to take the power plant out of the car (see Answer No. 32) in order to remove the magneto. Then remove crank case and transmission cover—take out the four cap screws that hold the flywheel to the crank shaft. You will then

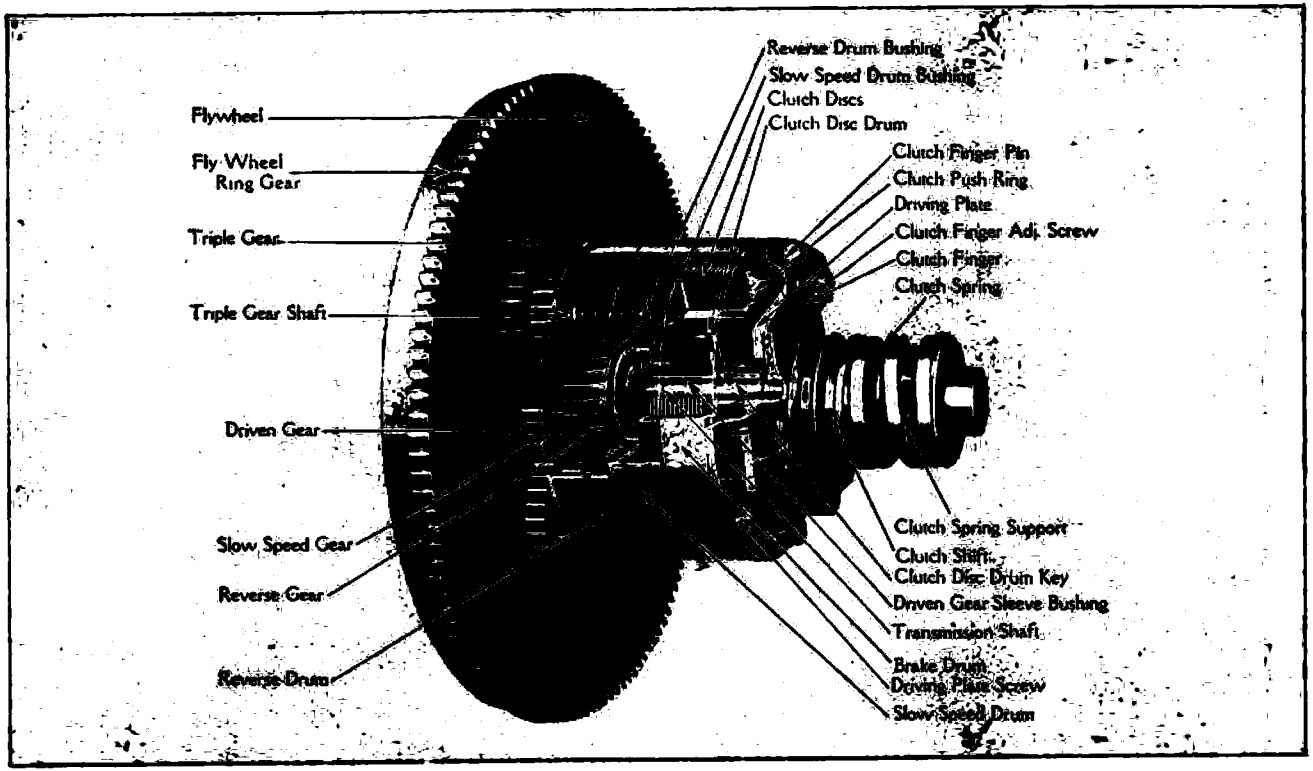


The Ford Magneto. The flywheel with magnets revolves while magneto coils remain stationary. (Cut No. 10)

have access to the magnets and entire magneto mechanism. In taking out these parts—or any parts of the car—the utmost care should be taken to make sure that the parts are so marked that they may be replaced properly.

When the Magneto gets out of order—what? **Answer No. 68**

The Ford magneto is made of permanent magnets and there is very little likelihood of their ever losing their strength, unless acted upon by some outside force. For instance, the attachment of a storage battery to the magneto terminal will demagnetize the magnets. If anything like this happens, it is not advisable to try to recharge them, but rather install a complete set of new magnets. The new magnets will be sent from the nearest agent or branch house, and will be placed on a board in identically the same manner as they should be when installed on the flywheel. Great care should be taken in assembling the magnets and lining up the magneto so that the faces of the magnets are separated from the surface of the coil spool just $\frac{1}{32}$ of an inch. To take out the old magnets, simply remove the cap screw and bronze screw which holds each in place. The magneto is often blamed when the trouble is a weak current caused by waste or other foreign matter accumulating under the contact spring, which is held in place by the binding post on top of the crank case cover. Remove the three screws which hold the binding post in place, remove binding post and spring and replace after foreign substance has been removed.



Transmission showing all gears in mesh. (Cut No. 11)